Course Outline

Bruce D. Marron

October 27, 2026

I. Introduction

The Modern World of AI, Satellites, and Drones

1. State of AI Capacity
2. State of Satellite Data
3. State of Drone Data

Using AI Professionally

1.

Use of Satellite Data in Ag

1.

Use of Drone Data in Ag

1.

II. Approaches

Systems Analysis

1. Fundamentals of Systems
   (a) Definitions
   (b) Systems Thinking
   (c) Systems Science
2. Fundamentals of Complexity
   (a) Definitions
   (b) Complex Adaptive Systems
   (c) Wicked Problems

DRAFT

3. Fundamentals of Scale
    (a) Spatial Scales
    (b) Temporal Scales

Data Analytics and Predictive Models

1. Probability and Statistical Models
2. Agent-Based Models
3. Imaginative Scenario Models

III. Topics

Standard Concepts and Frameworks of Sustainability

1. Standard Concepts of Sustainability
    (a) Brundtland Commission
    (b) UN Sustainable Development Goals
    (c) Marshall and Toffel

2. Standard Frameworks for Sustainability
    (a) Ecological Sustainability
    (b) Social Sustainability
    (c) Economic Sustainability
    (d) Agricultural Sustainability
3. Standard Concepts of Resilience
    (a) Robustness
    (b) Rebounding Capacity
    (c) Adaptability

Toward a Sustainable and Resilient Biosphere

1. Stewardship of the Global Climate System
    (a) Necessary and Sufficient Conditions
    (b) Best-Case Scenarios

DRAFT

2. Stewardship of Terrestrial Ecosystems
    (a) Necessary and Sufficient Conditions
    (b) Best-Case Scenarios
3. Stewardship of Aquatic Ecosystems
    (a) Necessary and Sufficient Conditions
    (b) Best-Case Scenarios

Toward a Sustainable and Resilient Civilization

1. Stewardship of Systems for Existential Human Needs
    (a) Necessary and Sufficient Conditions
    (b) Best-Case Scenarios
2. Stewardship of Systems for Higher Human Needs
    (a) Necessary and Sufficient Conditions
    (b) Best-Case Scenarios
3. Stewardship of Systems for Human Innovation
    (a) Necessary and Sufficient Conditions
    (b) Best-Case Scenarios

IV. Survival

At the Core

1. Your Cosmology and Worldview
    (a) Your Core Beliefs
    (b) Your Tolerance Levels
    (c) Your Imagination

Transition vs Collapse

1. Your Community
    (a) Who?
    (b) Why?
    (c) When?

DRAFT

   (d) Cooperation

2. Your Basic Needs
   (a) Water
   (b) Food
   (c) Shelter
   (d) Medicine
   (e) Energy
   (f) Security
   (g) Companionship and Community
3. Your Knowledge Base
4. Your Plan

DRAFT